

# RAG & Multi-Agent Application Manual Testing Guide

## Complete Manual Testing Guide for AI Agent & RAG System

This guide walks you step-by-step through manually testing every feature and module across all weeks using the test documents in this folder.

### ■ Test Documents in This Directory

| File Name                             | Best Used For                | What It Tests                                                                                                                      |
|---------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| customer_support_master_handbook.md   | Knowledge Base & General RAG | Error codes (ERR-4032, ERR-5001, AUTH-909), 30-day refund window, hardware returns, subscription tiers.                            |
| legal_commercial_master_agreement.md  | Week 6 Legal Contract Judge  | Limitation of liability (\$USD 12-month cap), indemnification obligations, 60-day termination for convenience, GDPR data deletion. |
| enterprise_sla_and_billing_policy.txt | Week 7 Loops & Week 10 Squad | 99.95% Platinum SLA uptime credit formulas, frontline \$100 refund authorization limits, payment retry intervals.                  |
| agent_security_injection_test_doc.md  | Week 8 Agent Security        | Benign documents with embedded indirect prompt injections and destructive tool call probes.                                        |

### ■ Step 0: Upload Documents into the Knowledge Base

- Open your browser at `http://localhost:5173`.
- Click **Knowledge Base** in the left sidebar navigation.
- Click the blue **Select & Upload File** button.
- Select `customer_support_master_handbook.md` (and optionally `legal_commercial_master_agreement.md`).
- The system will chunk and index the document into the local SQLite vector database automatically.

### ■ Step-by-Step Manual Test Scenarios by Week

#### 1. Live Chat Testing

- Go to **Live Chat** in the sidebar.
- Test queries:
  - `*"How do I fix database connection timeout error ERR-4032?"*`
    - **Expected:** Cites the resolution runbook (restart docker backend, clear browser cache, check port 8000).
  - `*"Can I return an opened hardware developer kit after 40 days?"*`
    - **Expected:** Informs user that physical hardware must be `*unopened*` in original packaging (even though the window is 45 days).

- **Expected:** Compares \$49/mo (50k calls) vs \$499/mo (unlimited calls, 99.95% SLA).

---

## 2. Week 4: RAG Retrieval & Chunk Debugger

- Go to **RAG Debugger (W4)** in the sidebar.
- Enter query: ERR-4032
- **What to observe:**
- **Query Rewriting & Tokens:** Extracts keyword token `err-4032` and exact error code.
- **Hybrid RRF Breakdown:** Inspects **BM25 Keyword score** (high for exact code) vs **Vector Cosine score**.
- **Failure Classifier:** Confirms SUCCESS classification.
- Enter query: Rocket launch date on Mars (Out-of-domain query)
- **What to observe:**
- Classifier detects RETRIEVAL\_FAILURE or NO\_MATCH and suggests ingestion remedy.
- Click **Run Benchmark** to view quantitative **Hit-Rate@3** and **MRR Score**.

---

## 3. Week 5: Error Analysis & Failure Taxonomy

- Go to **Error Analysis (W5)** in the sidebar.
- Click **Seed 20 Sample Traces** if empty.
- Click any trace from the left column:
- Read the User Query  $\rightarrow$  Retrieved Chunks  $\rightarrow$  LLM Answer.
- Test the **Hand-Coding Form**: Select Pass or Failure.
- Type an honest one-sentence note (e.g., "LLM missed the hardware unopened packaging requirement").
- Select Category and Severity (S = 1 to 5).
- Click **Save Annotation**.
- Switch to sub-tab **2. Ranked Error Taxonomy** to observe  $F \times S$  rankings update live.

---

## 4. Week 6: Legal Contract Clause Judge Validation

- Go to **Legal Judge (W6)** in the sidebar.
- Click **Run 26-Case Evaluation Suite**.
- **What to observe:**
- **Deterministic Assertions Split:** Tests regex rules, required keyword checks, and prohibited words before invoking LLM.
- **Few-Shot Disagreement Calibration:** Tests the LLM Judge on ambiguous contract clauses (e.g. mutual indemnification vs uncapped data breach exposure).
- Inspect pass rate, agreement percentage, and taxonomy breakdown.

---

## 5. Week 7: Agent Loops vs Fixed Workflows

- Go to **Agent Loops (W7)** in the sidebar.
- Click **Run Live Benchmark Race**.

- **What to observe:**
- Compares the **Fixed 3-Step Pipeline** (fast, rigid, fails on complex queries) vs the **ReAct Agent Loop** (dynamic multi-hop tool reasoning).
- Shows when the agent loop is worth the extra latency vs when a deterministic workflow wins.

---

## 6. Week 8: Agent Security & Failure Modes

- Go to **Agent Security (W8)** in the sidebar.
- Under **Prompt Injection Simulator**:
- Select **Indirect Prompt Injection**.
- Toggle **Defense Guardrails: OFF**  $\rightarrow$  Click **Simulate Attack**.
- Notice how the unmitigated agent follows the prompt injection override.
- Toggle **Defense Guardrails: ON**  $\rightarrow$  Click **Simulate Attack**.
- Notice how the sanitized agent rejects the injection and maintains task boundaries.
- Inspect the **Failure Mode Catalog** (Loop Trap, Wrong Tool, Made-Up Input, Quiet Surrender, Trajectory Gap).

---

## 7. Week 9: MCP Protocol Studio

- Go to **MCP Studio (W9)** in the sidebar.
- Under **Architecture**: Observe the Host  $\rightarrow$  Client  $\rightarrow$  Server JSON-RPC socket plumbing.
- Under **Connected Servers**:
- Click **Perform Handshake** to view the raw `initialize` JSON-RPC response.
- Under **Tools Discovery**:
- View tools exposed dynamically over `tools/list`.
- In **Tool Call Sandbox**, enter `{"order_id": "#90214"}` and click **Call Tool via MCP** to execute over `tools/call`.

---

## 8. Week 10: Multi-Agent & A2A Studio

- Go to **Multi-Agent (W10)** in the sidebar.
- Under **1. Single vs Multi-Agent Race**:
- Select Track: **A: Support**.
- Click preset ticket: `"Ticket #1: Technical Payment Gateway + Enterprise SLA Refund"`.
- Choose Worker Flow: **Parallel Workers**.
- Click **Run Benchmark Race**.
- **What to observe:**
- **Quality Score**: Compares Single Agent vs Squad.
- **Latency**: Measures wall-clock time.
- **Tokens Used**: Observe the **~7x token explosion** due to the Context Re-Send Tax!
- **Official Empirical Verdict**: Notice how the system objectively evaluates if the squad was worth the cost.
- Under **2. Squad Architecture & AgentCards**:
- Inspect the JSON **A2A AgentCards** (`support_triage_manager.json`, `tech_diagnostic_specialist.json`, `billing_sla_specialist.json`).

- Click **Simulate A2A Task Handshake** to watch the submitted  $\rightarrow$  working  $\rightarrow$  completed lifecycle.
- Under **3. Context Re-Send Cost Breakdown**:
- Review the visual diagram of context propagation.
- Under **4. MCP vs A2A & Frameworks**:
- Read the structured comparison table comparing MCP to A2A, and CrewAI to AutoGen.